

TEST OF MATHEMATICS FOR UNIVERSITY ADMISSION

PAPER 2: MATHEMATICAL REASONING

Practice paper · Set 10

75 minutes

INSTRUCTIONS TO CANDIDATES

Please read these instructions carefully, but do not open the question paper until you are told that you may do so.

A separate answer sheet is provided for this paper. You require a soft pencil and an eraser. Please complete the answer sheet with your candidate number, centre number, date of birth and full name.

This paper contains **20 questions**. For each question, choose the **one** answer you consider correct and record your choice on the separate answer sheet. If you make a mistake, erase thoroughly and try again.

There are no penalties for incorrect responses, only marks for correct answers, so you should attempt all 20 questions. Each question is worth one mark.

You can use the question paper for rough working or notes, but no extra paper is allowed.

You must complete the answer sheet within the time limit.

Calculators and dictionaries are NOT permitted.

There is no formulae booklet for this test.

Please wait to be told you may begin before turning this page.

This is a practice paper generated from a question bank written to the TMUA content specification. It is not an official UAT-UK paper. Answers and worked solutions are in the accompanying solutions booklet.

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- 1 Express $\frac{5}{3 + 2\sqrt{5}}$ in the form $a + b\sqrt{5}$ where a and b are rational.
- A** $15 - 10\sqrt{5}$
- B** $\frac{-15 + 10\sqrt{5}}{11}$
- C** $\frac{15 + 10\sqrt{5}}{29}$
- D** $\frac{15 - 10\sqrt{5}}{11}$
- E** $-15 + 10\sqrt{5}$
- F** $\frac{-15 + 10\sqrt{5}}{29}$

- 2 A line passes through $(1, 2)$ with gradient m . It crosses the x -axis at A and the y -axis at B , and the midpoint of AB is $(1, 2)$. Find m .

A -2

B $\frac{1}{2}$

C 2

D -4

E $-\frac{1}{2}$

F -1

3 Given that $\int_0^a (2x + 1) dx = 12$ where $a > 0$, find a .

A 4

B 12

C 2

D 3

E $\sqrt{12}$

F 6

4 Which of the following is the negation of ' $x > 3$ **and** $y \leq 5$ '?

A $x < 3$ and $y > 5$

B $x < 3$ or $y \geq 5$

C $x \leq 3$ or $y \geq 5$

D $x \leq 3$ and $y > 5$

E $x \leq 3$ or $y > 5$

- 5 The digits 1 to 9 are placed in a 3×3 grid so that every row and every column has the same sum. What is that sum?

- A 45
- B 15
- C 20
- D 18
- E 12
- F 10

6 Claim: for all real x , if $x^2 > 9$ then $x > 3$.

I Suppose $x^2 > 9$.

II Taking square roots of both sides, $x > 3$.

III This proves the claim.

Which of the following is true?

A The claim is false; the first error is on line II, since $x = -4$ satisfies $x^2 > 9$ but not $x > 3$.

B The claim and proof are both correct.

C The claim is true but line II should say $|x| > 3$ and then deduce $x > 3$.

D The claim is false; the first error is on line I.

E The claim is true; the first error is on line III.

- 7 Consider the statement: **if n is a multiple of 6, then n is a multiple of 3**. Which of the following is its contrapositive?
- A** If n is not a multiple of 3, then n is not a multiple of 6.
 - B** If n is a multiple of 3, then n is a multiple of 6.
 - C** If n is not a multiple of 6, then n is not a multiple of 3.
 - D** n is a multiple of 6 if and only if n is a multiple of 3.
 - E** If n is a multiple of 3, then n is not a multiple of 6.

- 8** A geometric series has first term $a \neq 0$ and common ratio r . P : ' $-1 < r < 1$ '. Q : 'the series has a sum to infinity'. Which one of the following is true?
- A** P is necessary and sufficient for Q .
 - B** P is necessary but not sufficient for Q .
 - C** P is sufficient but not necessary for Q .
 - D** P is neither necessary nor sufficient for Q .

- 9 For a positive integer n , consider the statement: **if n is prime, then n is odd**. Which one of the following is true?
- A The statement is true and its converse is true.
 - B The statement is true and its converse is false.
 - C The statement is false and its converse is true.
 - D The statement is false and its converse is false.

10 A 4×4 grid of squares is coloured so that every row and every column contains exactly two black squares. How many black squares are there in the grid?

A 6

B 16

C 10

D 8

E 4

F 12

11 Which of the following geometric series is/are convergent?

I $1 + 0.9 + 0.81 + \dots$

II $3 - 3 + 3 - 3 + \dots$

III $100 + 10 + 1 + 0.1 + \dots$

A none of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III

- 12** The circle C has centre (a, b) and radius r . P : ' $r > |b|$ '. Q : ' C crosses the x -axis at two distinct points'. Which one of the following is true?
- A** P is necessary and sufficient for Q .
 - B** P is necessary but not sufficient for Q .
 - C** P is sufficient but not necessary for Q .
 - D** P is neither necessary nor sufficient for Q .

- 13** For a positive integer n it is known that: (i) if n is a multiple of 4 then n is not a multiple of 6; (ii) n is a multiple of 12 or n is a multiple of 4. What can be deduced?
- A** The premises are inconsistent unless n is a multiple of 4 but not of 6; in particular n cannot be a multiple of 12.
- B** n is not a multiple of 4.
- C** n is a multiple of 12.
- D** n is a multiple of 6.
- E** Nothing can be deduced.

- 14** Let $f(x) = \sqrt{x-3}$ for $x \geq 3$ and $g(x) = x^2 + 3$. Which of the following statements is/are true?
- I** $g(f(x)) = x$ for all $x \geq 3$
 - II** $f(g(x)) = x$ for all real x
 - III** $f(g(x)) = |x|$ for all real x
- A** none of them
- B** I only
- C** II only
- D** III only
- E** I and II only
- F** I and III only
- G** II and III only
- H** I, II and III

- 15** Let $f(x) = x^2$ for all real x , and let g be the function whose graph is obtained by translating the graph of $y = f(x)$ by 2 units in the positive x -direction and then reflecting in the y -axis. Which of the following statements is/are true?

I $g(x) = (x + 2)^2$

II $g(x) = (x - 2)^2$

III The graph of g has its vertex at $(-2, 0)$

A none of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III

- 16** Alice, Ben and Cara each make one statement. Alice: 'Ben is lying.' Ben: 'Cara is lying.' Cara: 'Alice and Ben are both lying.' Exactly one of them is telling the truth. Who is it?
- A** Ben
 - B** Alice
 - C** Cara
 - D** It cannot be determined.
 - E** The situation is impossible.

17 A student writes the following argument.

I $0.1 > 0.01$.

II Taking logarithms to base 10 of both sides: $-1 > -2$.

III Multiplying both sides by $\log_{10} \frac{1}{2}$, which is negative, reverses the inequality:
 $-\log_{10} \frac{1}{2} < -2\log_{10} \frac{1}{2}$.

Which of the following is correct?

A All three lines are correct.

B Line II is wrong: taking logarithms reverses an inequality.

C Line III is wrong: $\log_{10} \frac{1}{2}$ is positive.

D Line II is wrong: $\log_{10} 0.1 = 1$.

E Line I is wrong.

18 A set of five distinct positive integers has median 10 and mean 10. Which of the following statements must be true?

- I** The largest value is at least 12.
- II** The smallest value is at most 8.
- III** The range is at least 4.

- A** none of them
- B** I only
- C** II only
- D** III only
- E** I and II only
- F** I and III only
- G** II and III only
- H** I, II and III

19 Every element of a set S of integers is either a multiple of 3 or a multiple of 5. Every multiple of 3 in S is negative. S contains the number 10. Which of the following statements must be true?

- I** S contains a positive multiple of 5.
- II** S contains no positive multiple of 3.
- III** S contains a negative number.

- A** none of them
- B** I only
- C** II only
- D** III only
- E** I and II only
- F** I and III only
- G** II and III only
- H** I, II and III

- 20** For how many integer values of n with $1 \leq n \leq 100$ is $\sqrt{n} + \sqrt{n+1}$ rational?
- A** 9
 - B** 1
 - C** 10
 - D** 0
 - E** 2
 - F** 100

END OF TEST